

Quarter I review paper · Algebra Grade 11

40 minutes 100 marks 14 questions + bonus No calculator

Lessons 1–25 · Limits, including limits at infinity · The derivative and the five rules · The chain rule · Modulus · Tangent and normal · Investigating a function · Extremum problems

- Show your working — method carries marks. A limit or a derivative written down with no working scores nothing.
- For a limit at infinity, divide numerator and denominator by the highest power of x and say which terms you are sending to zero.
- For a stationary point, state its nature and the test you used.

NAME _____ CLASS _____ DATE _____ MARK _____ / 100

A Section A · Five marks each

One or two lines of working is enough.

EASY

30 marks

1 Differentiate: $y = 4x^3 - 5x + 7$ [5]

2 Evaluate: $\lim_{x \rightarrow \infty} \frac{3x^2 + 2x}{x^2 - 5}$ [5]

3 Evaluate: $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$ [5]

4 For $y = x^2$, the value of x changes from 3 to 3.1. Find Δy , and the gradient of the secant through the two points. [5]

5 Differentiate: $y = (3x + 1)^5$ [5]

6 Solve: $|2x - 6| = 10$ [5]

B Section B · Eight marks each

Method marks are available even when the final answer is wrong.

MEDIUM

40 marks

7 Evaluate, showing the division in each case:

(a) $\lim_{x \rightarrow \infty} \frac{2x^2 - 3x + 1}{5x^2 + 4}$ [3]

(b) $\lim_{x \rightarrow \infty} \frac{x^2 + 1}{x^3 - 2}$ [2]

(c) Explain why $\lim_{x \rightarrow \infty} \frac{x^3 + 2x}{4x^2 - 1}$ does not exist. [3]

8 $f(x) = x^2 - 3x$.

(a) Show that $\frac{f(2+h) - f(2)}{h} = h + 1$. [5]

(b) Hence write down $f'(2)$. [3]

9 Differentiate $y = \frac{2x-1}{x^2+3}$ and find the gradient of the curve at $x = 1$. [8]

10 The curve $y = x^3 - 4x$. Find the equation of the tangent and the equation of the normal at the point where $x = 2$. [8]

11 $y = \sqrt{x^2 + 9}$. Find $\frac{dy}{dx}$ and its value at $x = 4$. [8]

C Section C • Ten marks each

Full method required. Define your variable before you differentiate.

HARD

30 marks

12 $y = x^3 - 6x^2 + 9x$.

(a) Find the stationary points and determine the nature of each. [6]

(b) State the intervals on which the function is increasing and decreasing, and sketch the curve.

[4]

- 13 A closed cylinder is to hold $128\pi \text{ cm}^3$. Find the radius that makes its total surface area least, and state that least area in terms of π . Show that your value gives a minimum. [10]

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- 14 (a) Evaluate $\lim_{x \rightarrow \infty} \frac{\sqrt{9x^2 + 1}}{2x + 5}$. [4]
(b) Evaluate $\lim_{x \rightarrow \infty} (\sqrt{4x^2 + 3x} - 2x)$. [6]

Bonus

- ★ Attempt this only when the rest is finished. It can make up marks lost elsewhere, but your total is still recorded out of 100. BONUS +10 marks

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- 15 $y = \frac{8x}{x^2 + 4}$ for $x \geq 0$.
(a) Show that the curve has a maximum at $x = 2$ and find its value. [6]
(b) Find $\lim_{x \rightarrow \infty} y$ and say what it tells you about the shape of the curve. [4]